

PRODUCTION OF ALLOYS

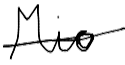
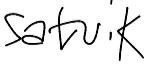
MIS41160 OPTIMISATION IN BUSINESS

Statement of Authorship

We, the undersigned members of this project group, hereby declare that:

1. This report is the result of our own work, except where otherwise stated and fully acknowledged through appropriate citations and references.
2. Each member has contributed meaningfully to the completion of this project.
3. We confirm that this submission complies with all University College Dublin (UCD) policies on academic integrity, plagiarism, and the use of generative artificial intelligence.

Team Members & Signatures

Student Name	Student Number	Signature	Date
Levi Kosisochukwu Simon-Eboh	24299675		28/11/2025
Inioluwa Makinde	25252861		28/11/2025
Mustafa Mohammed	25228596		28/11/2025
Satvik Kumar	25227818		28/11/2025

Team Nominee: Satvik Kumar

The team nominee is responsible for uploading this report and all associated files to Brightspace.

Table of Contents

<i>Executive Summary</i>	3
<i>Introduction and Policy Context</i>	3
<i>Literature Review:</i>	4
<i>Base Case Alloy Optimization Model</i>	5
<i>Extended Policy-Aware Model</i>	5
Modelling Assumptions.....	6
<i>Mathematical Formulation</i>	6
Decision Variables.....	7
Parameters.....	7
Objective Function.....	7
Constraints	7
Material balance constraints:	8
Composition constraints:	8
Availability constraints:.....	8
Demand constraints for the two markets:	8
<i>Tariff Index and Destination-Specific Costs</i>	8
Model Implementation in Mosel	8
Scenario-Based Sensitivity Analysis	9
Qualitative Results and Key Insights.....	9
<i>Conclusion</i>	11
<i>References</i>	11

Executive Summary

A policy-conscious optimisation model of steel alloy production was created. The primary challenge was to generalize the classical linear programming blending problem to reflect the modern carbon regulation and trade tariffs. The base model was used as a starting point. This report discusses the below three extensions:

- i. Destination-specific multi-market (European Union and United States) production.
- ii. Carbon emissions via a carbon price representing EU ETS/CBAM conditions.
- iii. Tariffs, which are based on the destination market and the origin of the material.

The objective minimises the total delivered cost which is the total cost of raw material, carbon cost and tariff cost. Constraints make sure that production balances, composition requirements, availability constraints, and fixed destination demands are met. Discussed are optional sustainability restrictions like a carbon cap and minimum content of recycled materials. To illustrate the way the model can be operationalized in practice, a Mosel file is also provided.

Qualitative analysis suggests that internalisation of carbon and tariff costs makes recycled and low-emission materials structurally more appealing, which tends to lower the total cost and policy exposure relative to a cost-only baseline. The primary contribution of this report is pedagogical: it demonstrates how the environmental and trade policies can be incorporated strictly into the known pattern of optimization of production in the process industries.

Introduction and Policy Context

The current policy environment in which regulatory and trade policies have a direct influence on the cost of production is currently facing steel producers who supply international markets. On the environmental front, the European Union has come up with the Carbon Border Adjustment Mechanism (CBAM) in addition to the current Emissions Trading System (ETS). CBAM is aimed at making imported goods subject to a similar cost of carbon as that imposed on EU producers, thus deterring leakage of carbon and establishing incentives to use production paths with lower carbon emissions. Practically, it implies that the embedded emissions in steel products are no longer cost-neutral and they should be factored in explicitly during the production planning.

Simultaneously, the United States, with the presidency of Donald Trump, has intensified its application of Section 232 tariffs on imported steel and aluminium on the grounds of national security and national protection of the domestic industry. Most of the trading partners have been subjected to tariff rates of about 50 percent on steel and aluminium, and some countries have been subjected to lower rates or exemptions under bilateral arrangements. These tariffs are imposed on the steel content of imported goods and greatly raise the effective cost of supplying the US market, especially to producers using high-emission, primary-production routes.

It is against this backdrop that the procurement and production choices made by steel companies should bear in mind not only the direct costs of the raw materials but also the costs that are brought

about by policy. The report concerns a stylised form of such a decision, in which a familiar alloy blending model is taken as a starting point and extended to include carbon and tariff factors.

Literature Review:

The classical models of alloy production planning in the steel industry are usually developed as a linear or mixed-integer linear programme that optimises the cost of raw materials and processing with respect to quality and capacity restrictions. A set of such applications is introduced by Gueret, Prins and Sevaux (2002) with Xpress-MP, showing how the blending and grade-control problems can be formulated and solved using linear constraints. Based on this, Bellabdaoui et al. (2006) construct a mixed-integer linear programming (MILP) model of production planning that relates alloy selection to sequencing and capacity decisions and emphasises the need to incorporate operational realities into otherwise stylised optimisation formulations.

During the last years, environmental concerns have become part of steel optimisation models. Kong et al. (2015) introduce a green MILP model to optimise the use of by-product gas in iron and steel facilities, where the emissions and energy recovery clearly form part of the objective function. Their results indicate that inclusion of environmental performance can alter the optimum operating regime as well as utilisation of by-product streams. Lee et al. (2023) take this point a step further by optimising the energy systems of iron and steel plants, including fuel options, by-products gases and electricity acquisitions. Both contributions justify the notion that environmental terms (energy, emissions) can be considered as extra cost items, maintaining linear or mixed-integer structure and having a material impact on optimal solutions.

On the policy front, the Carbon Border Adjustment Mechanism (CBAM) by the European Commission formalises the principle that the embedded emissions in imported steel should be subject to a comparable carbon charge as the EU producers (European Commission, 2025). Hasanbeigi, Springer and Chobthiangtham (2024) examine the probable effect of CBAM on world steel trade, stating that the intensity of emissions and the availability of low-carbon pathways (including scrap-based electric arc furnaces) will become a major factor of competitiveness. This is consistent with the outcomes of optimisation that indicate that an increase in carbon prices shifts solutions toward inputs that are less emitting or recycled.

The trade policy has a complementary but different impact. The recent US Section 232 actions have increased tariffs on imported steel and aluminium by 50 percent over 25 percent with the details of the implementation outlined in an official Presidential Proclamation (White House, 2025a) and summarised in a fact sheet (White House, 2025b). These sources validate that tariff rates are destination and origin specific and cover the content of steel and aluminium in imports. When combined with CBAM, they provide an environment where both the intensity of carbon and trade paths become sources of effective costs to exporters.

Collectively, these literatures indicates that alloy production planning in steel is naturally amenable to linear and mixed-integer optimisation (Gueret et al., 2002; Bellabdaoui et al., 2006), environmental objectives can be included as explicit cost or constraint terms without compromising

tractability (Kong et al., 2015; Lee et al., 2023), and modern policy tools like CBAM and Section 232 tariffs fundamentally change the cost structures of internationally traded steel (The current research expands these findings by integrating the carbon pricing and destination-specific tariff costs in a single linear programming model of selecting alloys and production that is market-dependent).

Base Case Alloy Optimization Model

The alloy production linear programming model is the base case. A steel producer must prepare a given amount of shipbuilding steel i.e. 500 tonnes by using several accessible raw materials i.e. iron, copper and aluminium alloys (Gueret, Prins and Sevaux, 2002). The raw materials are known to have a chemical composition, a maximum limit on the quantity available and a unit cost. The steel should be within the required percentages of important elements like carbon, copper and manganese. The decision variables in the base model are the tonnes of each raw material incorporated in the blend. The goal would be to reduce overall cost of raw materials. In line with this goal, the model imposes:

- a production balance constraint, which says that the total steel output is the sum of the inputs of raw materials.
- composition limits, which guarantee that the weighted mean of every chemical component is within prescribed minimum and maximum values.
- accessibility limitations of every raw material and,
- a demand constraint that the steel ordered must be manufactured at least.

While this model is mathematically correct, it implicitly presumes a neutral policy environment. It does not include any extra costs for emissions or trade. Consequently, the model fails to represent the economic impacts of climate policy and tariffs that have become the strategic choice in most steel markets.

Extended Policy-Aware Model

This study extends the base model in three dimensions to make the optimisation problem more representative of the present day conditions. Starting with several destination markets, namely, the European Union (EU) and the United States (US), which have their respective exogenous demand level. This generates a requirement to monitor the quantities of steel manufactured to every destination and the distribution of raw materials to markets. Second, factoring in carbon expenses by allocating an emissions factor to every raw material and utilizing a price on carbon that is comparable to EU ETS or CBAM standards. Third, imposing tariffs as destination-based mark-ups on the price of raw materials utilized in the production of steel in the US market. The most important modelling decision is not to change the cost-minimisation goal but to remain and change it to profit maximisation. Each market is assumed to have a fixed demand, which is a contractual obligation, but not discretionary sales. With this assumption, the economically relevant question is how to fulfil a set of orders with minimum total cost, when the carbon and tariff costs are internalised. This methodology keeps continuity with the base case but provides a more detailed representation of the policy effects.

Modelling Assumptions

A number of assumptions are made in order to maintain tractability and concentrate on the relationship between production choices and costs that are influenced by policy:

- Steel Co. is located in a non-EU and non-US location but operates in both markets, exporting 43% to US customers and 57% to EU customers. This ratio is denoted by α_{US} and α_{EU} .
- EU and US markets are fixed on demand which must be met completely. The model does not decide on what quantity to sell.
- The steel grade supplied to all destinations should have the same specifications of chemical composition.
- The prices of raw materials, emission rates and tariff rates are linear in quantities used; economies of scale and nonlinear pricing are assumed to not apply.
- It is a single period and static model, where there is no capacity expansion, inventory or dynamic investment decisions.
- The policy parameters (carbon price and tariff rates) are considered as exogenous inputs to the scenario, which remain constant over a single run of the optimization.
- The data of the base material of carbon, copper, manganese and availability of the seven raw materials is taken directly as the production of alloys case in Gueret, Prins and Sevaux, Applications of Optimization with Xpress-MP, Chapter 6, Table 6.2 (Gueret et al., 2002).
- The base cost prices mirrored the prices of the textbook alloy and a green premium imposed on material 3 in our extension (Gueret et al., 2002).
- The green premium of €110/t on material 3 is a more expensive, less-emitting "green" iron grade, which is set to be of the same order as the price premia of green-steel reported in recent decarbonization literature (IEA, 2020; World Steel Association, 2022; Wang et al., 2008; Richardson-Barlow et al., 2022).
- Tariff_rate parameters are stylized values representing exposure to U.S. Section 232 steel and aluminum tariffs (nominally 50%), based on recent presidential proclamations and fact sheets; slightly reduced rates (0.45–0.48) capture partial exemption or routing scenarios for sensitivity analysis (White House, 2025a; White House, 2025b).
- The parameter $r_{cbam} = 0.085$ converts emissions into a cost term using the prevailing EU ETS/CBAM carbon price of approximately €85/tCO₂ in November 2025, consistent with European Commission benchmark documentation and ICE Endex allowance price data (European Commission, 2025; ICE Endex, 2025).
- The market-share weights $\alpha_{US} = 0.43$ and $\alpha_{EU} = 0.57$ are demand shares that are a base scenario divided between U.S. and EU exposure; they are only used to weight revenue and policy costs across regions in the use case design.
- Lastly, the minimum demand of 500 tonnes is based on the initial shipbuilding steel order size used in the Gueret et al. base problem (Gueret et al., 2002).

Mathematical Formulation

In this section, the algebraic form of the extended model is described. The model takes into account a collection of raw materials $M = \{1, \dots, 7\}$. The decision variables are:

Decision Variables

The key decision variables will be as follows:

- $x_m \geq 0$: the quantity of raw material $m \in M$ used in the blend (tonnes)
- $Q \geq 0$: the total quantity of final steel produced (tonnes)

Parameters

The extended model uses the following parameters:

- c_m^{base} : base_cost(m) (€/t)
- p_m^{green} : green_premium(m) (€/t)
- a_m : availability(m) (maximum tonnes of material m)
- C_m : carbon(m) (% by mass)
- Cu_m : copper(m) (% by mass)
- Mn_m : manganese(m) (% by mass)
- τ_m : tariff_rate(m) (ad valorem tariff rate, fraction)
- e_m : co2_emissions(m) (kg CO_2 per tonne)
- r_{CBAM} : r_cbam (cost per kg CO_2 in model units)
- α_{US}, α_{EU} : alpha_us, alpha_eu (US/EU market shares, $\alpha_{US} + \alpha_{EU} = 1$)
- Q^{min} : min_demand (minimum required production)

The effective unit costs are established as follows based on:

- Base effective cost:

$$c_m^{\text{eff}} = c_m^{\text{base}} + p_m^{\text{green}} \quad \forall m \in M$$

- US tariff-inclusive cost:

$$c_m^{\text{US}} = c_m^{\text{eff}} (1 + \tau_m) \quad \forall m \in M$$

- EU carbon-inclusive cost:

$$c_m^{\text{EU}} = c_m^{\text{eff}} + e_m r_{\text{CBAM}} \quad \forall m \in M$$

- Blended cost (exposure to US/EU markets):

$$\tilde{c}_m = \alpha_{US} c_m^{\text{US}} + \alpha_{EU} c_m^{\text{EU}} \quad \forall m \in M$$

Objective Function

The objective is to reduce total delivered cost, which is the sum of raw material, carbon and tariff cost:

$$\min Z_{\text{cost}} = \sum_{m \in M} \tilde{c}_m x_m$$

Subject to the following constraints.

Constraints

The model has a number of constraints sets that guarantee physical feasibility and policy compliance. All constraints in the model are linear and all the decision variables are restricted to be non-negative.

Material balance constraints:

$$\sum_{m \in M} x_m = Q$$

The following equations are used to explain this: the quantity of the steel manufactured should be equal to the amount of raw materials that were used to manufacture the steel.

Composition constraints:

for each chemical component c (e.g. carbon, copper, manganese).

Chemical minimum specifications:

$$\sum_{m \in M} C_m x_m \geq 2.0 Q$$

$$\sum_{m \in M} Cu_m x_m \geq 0.4 Q$$

$$\sum_{m \in M} Mn_m x_m \geq 1.2 Q$$

Chemical maximum specifications:

$$\sum_{m \in M} C_m x_m \leq 3.0 Q$$

$$\sum_{m \in M} Cu_m x_m \leq 0.6 Q$$

$$\sum_{m \in M} Mn_m x_m \leq 1.65 Q$$

These inequalities guarantee that the ratio of component c in the end steel is within the acceptable range. They are determined by obtaining the total value of component c as a weighted sum and normalise it by the total amount produced.

Availability constraints:

Availability constraints for each raw material r :

$$x_m \leq a_m \quad \forall m \in M$$

These constraints ensure that the use of every raw material is not more than what is physically available.

Demand constraints for the two markets:

$$Q \geq Q^{\min}$$

These impose the condition that the model should generate sufficient steel to meet exogenous demand in every destination.

Tariff Index and Destination-Specific Costs

The tariff and carbon tax need a destination index since such destination specific costs do not depend on the mode of production but on the location of sale of the steel. The model can be used to calculate tariff on the US cost and the Carbon tax on the EU cost of steel sold to the respective destinations by indexing the cost in the model by destination.

Model Implementation in Mosel

The implementation of the optimization model is through FICO Xpress Mosel. The implementation is modular in order to increase clarity and reproducibility. The long-term policy-conscious formulation is specified in a central Mosel file, and the numerical values of the costs,

emission factors, availability constraints, demand levels and policy parameters are provided by external data files.

Scenario based sensitivity analysis is done with a separate Mosel script. In this script, the optimization model is repeatedly solved with various values of the policy parameters without changing the underlying constraint structure.

Scenario-Based Sensitivity Analysis

The sensitivity analysis involves the solution of the entire optimisation model in six stylised policy and market conditions. Scenario 1 is the calibrated baseline (43% US / 57% EU, existing CBAM and tariff rates). Scenario 2 increases the EU price of carbon twice, Scenario 3 increases the US tariffs twice, Scenario 4 and Scenario 5 expose the market to the EU (20% US / 80% EU) and the US (80% US / 20% EU) respectively, and Scenario 6 is a stress test of high CBAM and high tariffs. In both instances, the model is originally solved to minimise the overall cost to determine the baseline cost and the amount of emissions to produce, and it is again solved with further emissions-reduction targets (5%, 10 and 12) to determine the incremental cost of decarbonisation.

The cross-scenario analysis demonstrates that at the cost-optimal point, the fixed level of emissions will stay at 956,467 kg CO₂, yet the overall cost will increase to 231,439 BC in the stress test of high CBAM (+25.0%), to 209,579 BC in the stress test of high tariffs (+13.2%), and to 255,920 BC in the stress test of a combined stress test with high CBAM and high tariffs (+38.3% Restructuring of sales to the EU makes the baseline cost 190,703 BC ([?]3.0%), and a US-centric portfolio makes the baseline cost 176,082 BC ([?]4.9%). In all situations, the model finds feasible solutions when a 10 percent reduction of emissions is imposed, at very high marginal abatement costs: about EUR1,638/tCO₂ in the baseline, EUR1,590/tCO₂ in high CBAM, EUR1,924/tCO₂ in high tariffs, EUR1,466/tCO₂ in the EU-focused case, EUR1,916/tCO₂ in the US-focused case, and EUR1,876/tCO. Making a 12% cut makes the problem unsolvable in all cases, which proves that the upper limit of abatement is not determined by policy parameters only but structural and chemical constraints. In general, the sensitivity analysis shows that the current EU carbon policy is more cost-effective than tariffs, market mix can slightly reduce or increase the effect, and decarbonisation of the existing process is technically constrained and costly.

Qualitative Results and Key Insights

Despite the fact that the emphasis of this report is on the development of a model, as opposed to comprehensive numerical analysis, the results of the scenario demonstrate some strong qualitative trends. In all six policy scenarios, the optimal cost solution remains primarily based on primary iron (80%) and secondary alloy (18.8%), with the same amount of emissions of 956,467 kg CO₂ and an intensity of 1,913 kg CO₂/t. Policy shocks act not by switching technology, but by raising the cost: doubling the EU carbon price adds to the baseline costs by approximately 25 percent, doubling US tariffs adds to costs by 13 percent and a combined stress test (high CBAM and high tariffs) adds to costs by 38 percent, but does not change the underlying material mix. Market geography is also important: EU-oriented sales mix (20% US / 80% EU) is approximately 3% more costly than the base, and the US-oriented mix (80% US / 20% EU) is nearly 5% cheaper. But with explicit targets of emissions-reduction (e.g. 5-10% reductions), the model demonstrates that

decarbonisation can be achieved, but it is very costly, with marginal abatement costs in the range of EUR1,460-EUR1,925 per tonne of CO₂ to achieve a 10% reduction. To achieve cuts more than 10-12, the optimisation problem is infeasible due to binding chemical and availability constraints, and this is a hard technical limit to what is possible with the available material set.

As a result, three lessons can be learned. One, carbon and tariff policies may significantly raise the cost of production (up to +38% in the stress scenario) without altering the physical process; thus, the firms require clear policy resilience planning, not process efficiency. Second, the transformation of market exposure between US-bound and EU-bound sales may alter costs by nearly 8 percentage points, indicating that destination strategy (where to sell) is nearly as significant as sourcing strategy (where to purchase). Third, despite the high policy incentives, the existing process has very high marginal abatement costs and very high feasibility constraints; significant decarbonisation will not be possible without optimisation of the current blend, but also material specifications or technology.

Conclusion

The report has generalised a classical alloy-blending optimisation problem to include carbon regulation, destination-specific trade tariffs and market-mix effects in a single linear programming. The model fills the gap between the traditional process-industry optimisation and the policy-constrained environment contemporary steel producers have to work in by enriching the cost structure with carbon and tariff elements and by adding US/EU exposure weights.

Policy shocks have the potential to raise the total cost by 13-38% with no change in the underlying material mix, whereas changes in market focus can change costs by nearly 8 percentage points, as indicated by the scenario analysis. At explicit emissions-reduction levels, the model provides viable solutions to a maximum of about 10% reduction, but at marginal abatement costs many times higher than current carbon prices and infeasibility above 10-12% indicating hard technical constraints in the current process.

These results suggest that quantitative optimisation is not only useful in cost minimisation, but also in diagnosing the location of policy and technical constraints at the point of binding. Carbon pricing and tariffs play a major role in the current case in terms of cost exposure and market policy, but not alone would prompt a clean technology switch; massive decarbonisation would need other material specifications, process routes or product design. To practitioners, the model offers a systematic method of comparing policy situations, evaluating the cost of marginal cuts in emissions, and determining when additional improvement requires an engineering breakthrough, as opposed to engineering efficiency of current inputs.

References

Bellabdaoui, A., Teghem, J., Loukil, T., Saadani, N. and Amodeo, L. (2006) 'A mixed-integer linear programming model for production planning', *International Journal of Production Research*, 44(17), pp. 3457–3473.

European Commission (2025) *Carbon Border Adjustment Mechanism*. Available at: taxation-customs.ec.europa.eu/carbon-border-adjustment-mechanism_en (Accessed: 26 October 2025).

Guéret, C., Prins, C. and Sevaux, M. (2002) *Applications of Optimization with Xpress-MP*. London: Dash Optimization.

Hasanbeigi, A., Springer, C. and Chobthiangtham, P. (2024) *The impact of the EU CBAM on global steel trade*. San Francisco: Global Efficiency Intelligence.

ICE Endex (2025) EU ETS Carbon Allowance Prices, November 2025. Amsterdam: ICE Endex.

International Energy Agency (2020) *Iron and Steel Technology Roadmap*. Paris: IEA. Available at: <https://doi.org/10.1787/3dcc2a1b-en>.

Kong, H., Li, H., Wang, Y., Zhang, Y. and Zheng, Q. (2015) 'A green mixed integer linear programming model for optimisation of byproduct gases in iron and steel industry', *Journal of Iron and Steel Research, International*, 22(9), pp. 795–803.

Lee, S.Y., Park, J., Kim, H. and Choi, Y. (2023) 'Optimization of iron and steel manufacturing plant energy systems', *IET Generation, Transmission & Distribution*, 17(18), pp. 5027–5040.

Richardson-Barlow, C., Chalmers, H., Gross, R. and Heptonstall, P. (2022) 'Policy and pricing barriers to steel industry decarbonisation', *Energy Policy*, 168, 113100.

Wang, K., Wang, C., Lu, X. and Chen, J. (2008) 'A model on CO₂ emission reduction in integrated steel plants', *Energy Policy*, 36(7), pp. 2567–2577.

White House (2025a) 'Adjusting imports of aluminum and steel into the United States', *Presidential Proclamation*, 2 June. Available at: <https://www.whitehouse.gov/presidential-actions/2025/06/adjusting-imports-of-aluminum-and-steel-into-the-united-states/>

White House (2025b) 'Fact Sheet: President Donald J. Trump increases Section 232 tariffs on steel and aluminum', *Fact Sheet*, 2 June. Available at: <https://www.whitehouse.gov/fact-sheets/2025/06/fact-sheet-president-donald-j-trump-increases-section-232-tariffs-on-steel-and-al>

World Steel Association (2022) *Steel's Contribution to a Low Carbon Future and the Role of the Circular Economy*. Brussels: World Steel Association.

APPENDIX:

Team Individual Reflections and Contributions

Inioluwa Makinde

My background was primarily in finance, and I had limited exposure to optimisation modelling or environmental policy frameworks. My understanding of carbon pricing and trade tariffs was insignificant. I viewed them as separate policy instruments rather than integrated dimensions of competitive cost structures.

My contribution to this project involved conducting the comprehensive literature review and calibrating all numerical parameters to reflect real-world conditions. Specifically, I researched and synthesized literature on classical alloy optimization models, green steel production economics, Carbon Border Adjustment Mechanism (CBAM), and US Section 232 tariff implementation. The main challenge I faced was validating data sources and ensuring parameter consistency. Emission factors varied across publications depending on system boundaries, production route, and regional assumptions. I resolved this by establishing explicit definitions and cross-validating calibrations across three independent sources.

Levi Simon-Eboh

Before this module, my understanding of linear programming was largely theoretical, limited to textbook and hypothetical examples with simple cost-minimization objectives and basic constraints. I had never encountered multi-dimensional optimization problems that incorporated real-world policy components.

My primary contribution was developing the mathematical formulation for the extended policy-aware model. Specifically, I was responsible for designing the destination-indexed decision variables, formulating the composite objective function that integrates raw material costs, carbon costs, and destination-specific tariff costs, and translating these into rigorous algebraic notation for Section 4 of the report. I also worked closely with the implementation team (Satvik Kumar) to ensure that the mathematical structure preserved linearity while accommodating the complexity of multi-market, policy scenarios.

Satvik Kumar

Before taking this module, my experience with optimization was limited to basic solver use in Excel and a single undergraduate course on algorithms. I had never programmed in a specialized optimization language, and I was excited to implement a complex multi-constraint model.

My primary responsibility was implementing the extended optimization model in FICO Xpress Mosel and developing the scenario analysis framework. I translated the mathematical formulation into executable code. I also structured external data files to separate parameters from the core model logic, enabling efficient sensitivity analysis without code modification. Additionally, I designed and implemented the two numerical scenarios (baseline with zero carbon prices and tariffs; policy scenario with €85/tCO₂ carbon pricing and 50% tariffs) and wrote auxiliary Mosel scripts to extract and report results across scenarios. This assignment enhanced my technical and problem-solving capabilities. I gained practical proficiency with Mosel syntax, including advanced features such as loops for parameterized constraint generation, dynamic indexing to handle destination-specific variables, and data-read functions for external parameter integration.

Mustafa Mohamed

Before this module my analytical background was limited in business strategy rather than quantitative optimization. I was confident in qualitative reasoning and stakeholder analysis but had limited exposure to interpreting optimization outputs or understanding how numerical solutions translate into actionable insights.

My contributions to this project involved conducting qualitative analysis of the scenarios, synthesizing scenario results into managerial actions, and integrating all components into a cohesive report narrative. Specifically, I analysed how total cost, emissions, and material allocation changed between the baseline and policy scenarios, formulating the mechanisms by which carbon and tariff costs reshape sourcing strategies. Finally, I compiled the overall report structure and ensured that mathematical formulation, numerical parameters, and model were logically connected and understandable. The key challenge I faced was translating numerical outputs into clear language without oversimplifying. Through feedback and revision, I learned to lead with insights before explaining the underlying mechanisms.